



SIMATS
ENGINEERING



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SIMATS ENGINEERING

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS | COURSE CODE: MLA0104

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STRUCTURE OF THE REPORT (DELIVERABLES 1–14)

1. Title of AI Application

Unitree G1 AI-Powered Generalist Humanoid Robot System (2026 Hardware Release)

Domain: Robotics (Scenario : Intelligent Robots for Industrial, Service, and Academic Applications)

2. Domain Selection

Selected Domain: Robotics (Domain 5)

Sub-domain Focus: Physical AI, Bipedal Humanoid Locomotion, Edge-AI Compute, and Dexterous Force-Feedback Manipulation.

3. Introduction and Background

The Unitree G1 represents a high-water mark in commercial physical AI and humanoid robotics in 2026. Standing at 1.32 meters tall and weighing approximately 35 kg, the robot features 23 to 43 degrees of freedom (DoF) driven by integrated high-torque permanent-magnet synchronous motors (PMSM) providing up to 120 N·m peak joint torque. The platform utilizes onboard edge-AI acceleration via the NVIDIA Jetson AGX Orin / Thor processor, executing large multimodal generative models (Unifolm world models) in tandem with real-time Deep Reinforcement Learning (DRL) policies. This architectural combination enables dynamic terrain navigation (>2.0 m/s), full-body collision recovery, voice instruction parsing, and multi-finger force-sensitive object handling.

4. Problem Identification

Traditional industrial robotics and dynamic automation systems face severe operational bottlenecks in human-centric workspaces:

- **Unstructured Environment Incompatibility:** Fixed robotic arms and automated guided vehicles (AGVs) cannot navigate human infrastructure such as staircases, narrow thresholds, or debris-laden floors.
- **Lack of Dexterous Manipulation:** Conventional parallel grippers lack multi-point tactile sensing and adaptive compliance, preventing them from handling soft, irregular, or delicate objects.
- **High Reconfiguration & Deployment Overhead:** Reprogramming classical automated systems for new assembly or logistics tasks demands prohibitive manual coding, safety enclosures, and physical layout changes.

5. Need for AI Solution

Classical control schemes relying on pre-computed Zero Moment Point (ZMP) trajectories fail when encountering unpredictable external forces or terrain changes. AI integration overcomes these limitations:

- **High-Dimensional State-Space Optimization:** Deep Reinforcement Learning (DRL) policies solve dynamic balance, inverse kinematics, and contact-wrench optimization equations at 500 Hz.
- **Zero-Shot Task Generalization:** Vision-Language-Action (VLA) foundation models parse unstructured verbal and visual commands into continuous motor control vectors without hand-coded state machines.
- **Force-Position Hybrid Compliance:** Neural networks adjust joint stiffness dynamically in response to unexpected collisions, ensuring human safety and robot stability.

6. AI Technologies Used

The G1 humanoid platform relies on an integrated multi-tiered physical AI framework:

- **Deep Reinforcement Learning (DRL):** Trained in massively parallel physics simulators (NVIDIA Isaac Sim) using sim-to-real domain randomization for robust gait synthesis.
- **Multimodal Vision-Language Models (VLM/LLM):** Generative UnifolM models process multi-camera RGB streams and audio commands to execute multi-step object manipulation plans.
- **3D Spatial Perception & Point-Cloud SLAM:** Fuses point clouds from 3D LiDAR and stereo depth cameras to generate volumetric voxel occupancy maps for dynamic path planning.
- **Imitation & Teleoperation Learning:** Maps human motion vectors captured through VR tracking and haptic gloves directly into the robot's joint torque space.

7. Working Mechanism

The system executes an ongoing perception-action control loop operating at multiple frequencies:

1. Multi-Sensor Perception (60–300 Hz): The Livox MID-360 LiDAR constructs a 360° point cloud while the Intel RealSense D435i camera tracks near-range obstacle geometry. Ground force sensors measure foot impact.

2. State Estimation & Filtering (1000 Hz): An Extended Kalman Filter (EKF) fuses torso 6-axis IMU readings with dual-motor absolute encoders to track center-of-mass (CoM) state dynamics.

3. Neural Policy Execution (500 Hz): The DRL controller running on the NVIDIA Jetson Orin/Thor module evaluates the state vector and calculates target joint angles and torque limits every 2 ms.

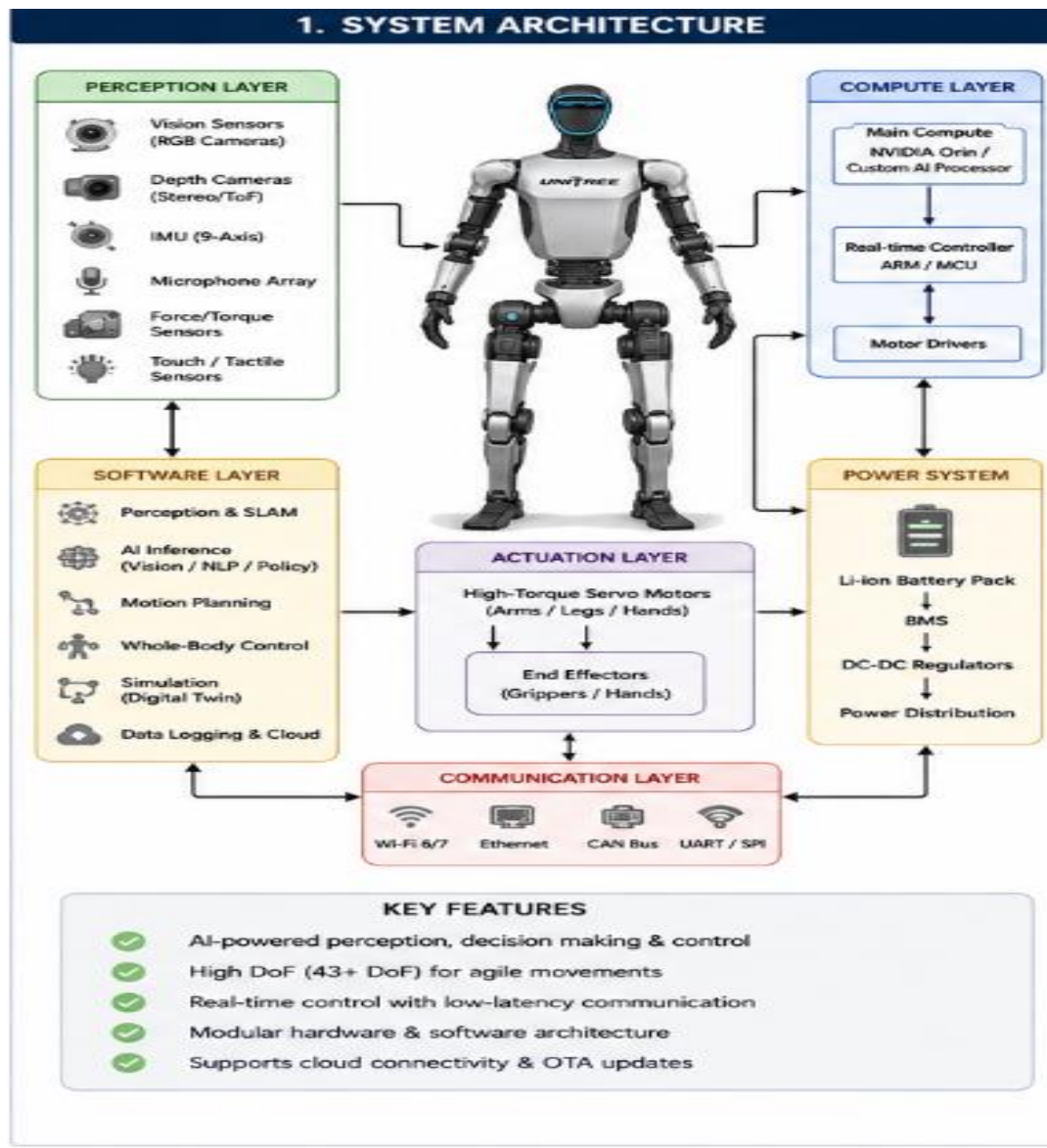
4. Low-Level Motor Control (10,000 Hz Internal FOC): Dedicated PMSM motor drivers execute field-oriented current loops over a real-time CAN-FD / DDS bus to actuate joint positions.

8. Data Sources and Processing

The robot processes diverse heterogeneous data streams required for real-time autonomy:

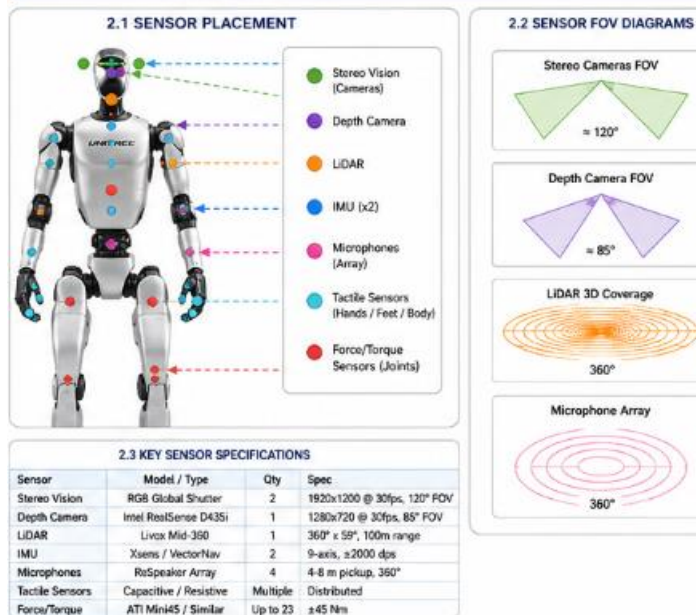
- **Visual & Spatial Point Clouds:** LiDAR spatial matrices (~200,000 points/sec) and RGB-D depth channels (1080p @ 60 FPS) filtered into voxel grids.
- **Inertial & Proprioceptive Data:** 6-Axis IMU accelerations/angular velocities (1000 Hz) and dual absolute joint position encoders.
- **Tactile Force Feedback:** 17-zone capacitive tactile arrays per hand sampling pressure vectors to prevent slippage and monitor gripping force.

SYSTEM ARCHITECTURE



SENSOR DIAGRAMS

2. SENSOR DIAGRAMS



MATLAB WORKFLOW

3. MATLAB WORKFLOW



9. Real-World Implementation

The Unitree G1 platform is deployed across industrial, research, and logistics environments:

- **Academic Research Institutions:** Deployed at robotics labs (e.g., MIT, Stanford, UT Austin) for human-robot interaction and agile bipedal locomotion studies.
- **Smart Manufacturing & Assembly:** Integrated into industrial testbeds for fetching components, tool holding, and visual quality inspection.
- **Logistics & Warehouse Operations:** Utilized alongside Autonomous Mobile Robots (AMRs) for sorting packages and automated bin picking.

10. Impact and Benefits

- **High Torque Density:** Delivers up to 120 N·m peak torque on a lightweight 35 kg frame, enabling dynamic agility and high payload-to-weight performance.
- **Dynamic Impact Recovery:** Backdrivable planetary joint actuators allow the robot to maintain balance when impacted or push-tested.
- **Cost Democratization:** Significantly lowers entry costs compared to legacy research humanoids, accelerating research and deployment.

11. Challenges and Limitations

- **Thermal Dissipation:** Sustained maximum torque output causes motor heating, requiring active thermal management and torque throttling.
- **Battery Operational Bounds:** Onboard Li-ion battery capacity limits continuous high-speed movement to roughly 1 to 2 hours per charge.
- **Cybersecurity Vulnerabilities:** Real-time DDS middleware and wireless communication links require encryption against unauthorized command injection.

12. Future Scope and Emerging Trends

Emerging developments for next-generation humanoid platforms include:

- **Next-Generation Physical AI Hardware:** Transitioning to Blackwell-architecture NVIDIA Jetson AGX Thor modules capable of running 2000+ TFLOPS onboard.
- **Full-Body Artificial Tactile Skin:** Extending capacitive pressure-sensing skins across the entire arm and torso for safe collaborative human interaction.
- **Solid-State Battery Integration:** Adopting higher density battery chemistries to extend active operational time past 4 hours.

13. Conclusion

The Unitree G1 AI Humanoid Robot system represents a substantial advance in physical AI and robotics hardware. By fusing Deep Reinforcement Learning locomotion policies with multimodal world models, high-torque PMSM motors, and multi-sensor hardware architecture, the platform bridges the gap between structured factory automation and flexible general-purpose robotics. Through model-based control platforms like MATLAB/Simulink combined with real-time ROS 2 edge compute, humanoids are well-positioned to become practical collaborators across industrial, logistics, and research domains.

14. References

1. Unitree Robotics, 'Unitree G1 Humanoid Robot Hardware & SDK Technical Documentation', 2025–2026.
2. NVIDIA Corporation, 'NVIDIA Jetson AGX Thor & Isaac GR00T Architecture Technical Report', 2025–2026.
3. TrendForce, 'Global Humanoid Robotics Hardware & Edge AI Market Analysis', 2025–2026.

4. MathWorks, '*Robotics System Toolbox & ROS 2 Integration Workflows for Humanoid Control*', MATLAB Documentation, 2026.